

Engineering challenges in repo 1.0

#	Engineering Challenge	Core Engineering Friction / Issue	Identity Verification / Document Intelligence Connection	Compliance / Transaction Monitoring Connection
1	Identity-to-Transaction Entity Linkage	Customer, case, document, account and transaction identifiers may not align cleanly, creating false joins or missed associations.	Resolves identity evidence from multiple documents into a reliable customer/entity representation.	Ensures monitored transactions are attributed to the correct customer, account and compliance case.
2	Document-Intelligence Uncertainty Propagation	OCR confidence, poor image quality, extraction ambiguity and classification uncertainty are often lost after parsing.	Preserves confidence, provenance, bounding evidence and extraction-quality signals from source documents.	Downstream monitoring can increase scrutiny when identity evidence is weak rather than treating uncertain KYC as verified fact.
3	Cross-Document Identity Resolution	Names, initials, transliteration, punctuation, OCR corruption, address variations and inconsistent DOBs create contradictory identity records.	Reconciles Passport, PAN, Aadhaar, DL or other synthetic document evidence into a normalized identity.	Prevents fragmented customer profiles from hiding aggregate transactional risk or generating duplicate monitoring cases.
4	KYC Lifecycle Drift vs Live Transactions	Customer activity continues even when identity evidence becomes stale, expired, rejected or materially changed.	Tracks document expiry, verification status, refresh date and unresolved identity exceptions.	Transaction monitoring must incorporate KYC freshness and trigger enhanced review where activity continues against stale identity evidence.
5	Expected-Activity Profile Normalization	Occupation, business type, income, expected turnover and geographic information often exist as inconsistent text rather than computable attributes.	Extracts and normalizes customer-profile evidence from onboarding documents and declarations.	Converts KYC evidence into an expected-behaviour baseline for detecting abnormal volume, value, geography or counterparties.
6	Transaction-Hook Idempotency	Streaming systems may redeliver the same event, causing duplicated alerts, cases or risk-score inflation.	Maintains stable customer/account correlation keys originating from identity resolution.	Uses event IDs, deduplication and idempotent processing so duplicate transaction hooks do not create false compliance signals.
7	Out-of-Order and Late Events	Events may arrive after newer transactions or significantly after their actual business timestamp.	Identity and KYC context may also change over time, requiring the correct historical state to be reconstructed.	Temporal rules must use event time, windows and historical customer context rather than simplistic ingestion order.
8	Structuring / Threshold Avoidance	Individual transactions may appear harmless while the aggregate pattern indicates deliberate threshold avoidance.	Verified ownership and linked-account identity are necessary to aggregate transactions across the	Windowed aggregation identifies multiple sub-threshold transactions whose combined behaviour

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9	Rapid Transaction Velocity	Risk emerges from a sequence of events rather than from any single transaction.	correct customer relationship. Reliable identity linkage prevents activity from being incorrectly split across aliases or duplicate customer records.	constitutes a monitoring pattern. Detects bursts of transactions, many counterparties, rapid transfers or unusual frequency within configurable time windows.
10	High-Risk Corridor × Identity-Context Fusion	Geography by itself produces excessive false positives; risk depends on customer context and identity evidence.	Provides residency, nationality, address, identity confidence, KYC refresh state and unresolved-document discrepancies.	Fuses corridor/ geographic signals with customer identity context to produce more defensible alerts and prioritization.
11	Pass-Through / Funnel-Account Behaviour	Suspicious behaviour may involve rapid movement of incoming funds rather than unusually large individual transactions.	Establishes who actually owns or controls the account and links associated identity relationships.	Detects patterns such as incoming credit followed quickly by near-equivalent outward transfers or dispersal to multiple beneficiaries.
12	False-Positive Reduction vs Explainability	More sophisticated scoring may reduce alerts but can make decisions opaque and difficult to defend.	Retains source-document evidence, confidence, identity discrepancies and provenance.	Every alert links transactions, rule/pattern matches, KYC factors, policy versions and reasons so investigators can understand why it fired.
13	HITL Workflow Integrity	Human review can become an uncontrolled override channel if authority, state transitions and evidence changes are not governed.	Reviewers can inspect identity evidence, correct extracted fields and resolve document conflicts without silently rewriting history.	Alert disposition, escalation, suppression and override decisions are role-controlled, timestamped and auditable.
14	Policy / Rule / Model Version Drift	A decision made today may be impossible to reproduce later after thresholds, rules, prompts or models change.	Extraction/model/provider versions and identity-resolution policies must be preserved with each decision.	Monitoring rules, thresholds, risk weights and policy versions are recorded so historical alerts can be replayed and explained.
15	Evals, Observability & Production Readiness	Passing unit tests does not prove operational safety, monitoring quality, resilience or compliance readiness.	Evaluates extraction accuracy, identity resolution, contradiction handling, privacy, security and evidence lineage.	Evaluates alert precision/recall, replayability, latency, duplicate handling, temporal correctness, security, auditability, failure recovery and release gates.